

Session 8: FunctionsTopics:Declaration, Definition, Calling.Pass-by-value vs pass-by-reference.

Tasks

1) Declare a function called `getUserInitials` that takes a user's full name (like 'Virat Kohli') and returns their initials in uppercase (e.g., 'VK'). Call this function with your favorite cricketer's name and print the result.

Code-

```
#include <stdio.h>

#include <ctype.h>

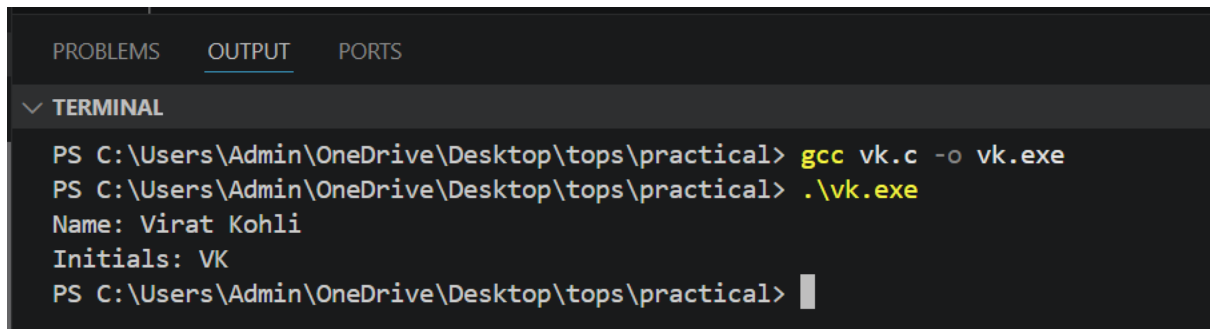

// Function to get initials
void getUserInitials(char name[]) {
    printf("%c", toupper(name[0]));

    // Find the first character after a space
    for (int i = 0; name[i] != '\0'; i++) {
        if (name[i] == ' ' && name[i + 1] != '\0') {
            printf("%c", toupper(name[i + 1]));
            break;
        }
    }
}


int main() {
    char name[] = "Virat Kohli";
```

```
printf("Name: %s\n", name);  
printf("Initials: ");  
getUserInitials(name);  
  
return 0;  
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS  
✓ TERMINAL  
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc vk.c -o vk.exe  
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\vk.exe  
Name: Virat Kohli  
Initials: VK  
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> |
```

**2) Create a function addToCart that takes a shopping cart array and a product name, adds the product to the cart, and prints the updated cart. Demonstrate how passing the cart array by reference allows changes to persist outside the function.

Hint: In languages like JavaScript, arrays are passed by reference. In C/C++, use pointers for reference behavior.**

Code-

```
#include <stdio.h>

#include <string.h>

#define MAX_PRODUCTS 10
#define MAX_NAME_LENGTH 50

// Function to add a product to the cart
void addToCart(char cart[][MAX_NAME_LENGTH], int *count, char product[]) {
    if (*count < MAX_PRODUCTS) {
        strcpy(cart[*count], product);
        (*count)++;

        printf("\nUpdated Cart:\n");
        for (int i = 0; i < *count; i++) {
            printf("%d. %s\n", i + 1, cart[i]);
        }
    } else {
        printf("Cart is full!\n");
    }
}
```

```
int main() {  
    char cart[MAX_PRODUCTS][MAX_NAME_LENGTH];  
    int count = 0;  
  
    // Add products  
    addToCart(cart, &count, "Laptop");  
    addToCart(cart, &count, "Mouse");  
    addToCart(cart, &count, "Keyboard");  
  
    // Print cart from main to show changes persist  
    printf("\nCart in main():\n");  
    for (int i = 0; i < count; i++) {  
        printf("%d. %s\n", i + 1, cart[i]);  
    }  
  
    return 0;  
}
```

Output:

```
✓ TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc product1.c -o product1.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\product1.exe

Updated Cart:
1. Laptop

Updated Cart:
1. Laptop
2. Mouse

Updated Cart:
1. Laptop
2. Mouse
3. Keyboard

Cart in main():
1. Laptop
2. Mouse
3. Keyboard
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> |
```

4) Write two functions: increase Followers By Value and increase Followers By Reference. Each should take a followers count and add 1000 to it, but one should use pass-by-value and the other pass-by-reference. Show how the original followers count changes (or doesn't) after calling each function.

Code-

```
#include <stdio.h>
```

```
void increaseFollowersByValue(int followers) {  
    followers = followers + 1000;  
    printf("Followers inside increaseFollowersByValue(): %d\n", followers);  
}
```

```
void increaseFollowersByReference(int *followers) {  
    *followers = *followers + 1000;  
    printf("Followers inside increaseFollowersByReference(): %d\n", *followers);  
}
```

```
int main() {  
    int followers = 5000;  
  
    printf("Original followers: %d\n\n", followers);  
  
    increaseFollowersByValue(followers);  
    printf("Followers after increaseFollowersByValue(): %d\n\n", followers);  
  
    increaseFollowersByReference(&followers);
```

```
printf("Followers after increaseFollowersByReference(): %d\n", followers);
```

```
return 0;
```

```
}
```

Output:

```
✓ TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc callby.c -o callby.exe
cc1.exe: fatal error: callby.c: No such file or directory
compilation terminated.
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc callby1.c -o callby1.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\callby1.exe
Original followers: 5000

Followers inside increaseFollowersByValue(): 6000
Followers after increaseFollowersByValue(): 5000

Followers inside increaseFollowersByReference(): 6000
Followers after increaseFollowersByReference(): 6000
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> █
```

4) Build a function format Price that takes a price in rupees and returns a string formatted like Flipkart's price tag (e.g., '₹1,599'). Use this function to display prices for three different products.

Code-

```
#include <stdio.h>

void formatPrice(int price) {
    if (price >= 1000)
        printf("₹%d,%03d", price / 1000, price % 1000);
    else
        printf("₹%d", price);
}

int main() {
    char product1[] = "Wireless Mouse";
    char product2[] = "Keyboard";
    char product3[] = "Headphones";

    int price1 = 1599;
    int price2 = 899;
    int price3 = 2499;

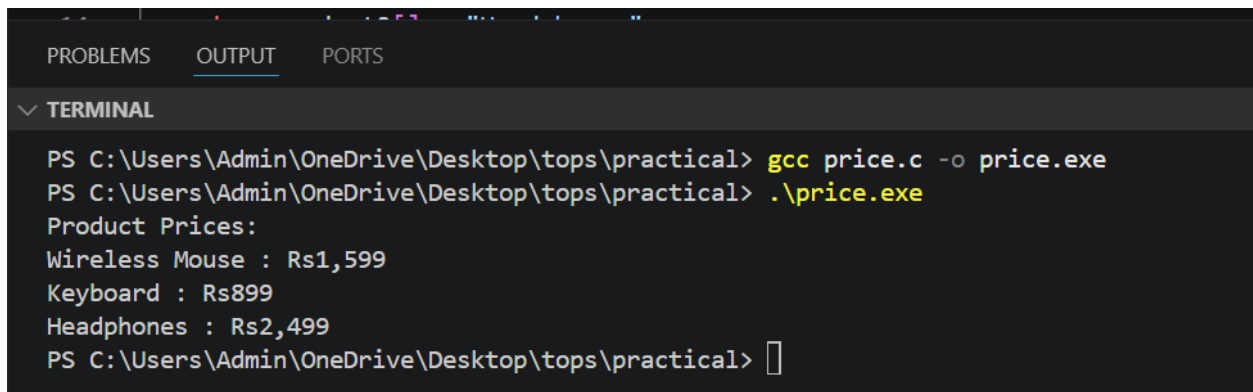
    printf("Product Prices:\n");

    printf("%s : ", product1);
```



```
formatPrice(price1);  
  
printf("\n");  
  
printf("%s : ", product2);  
formatPrice(price2);  
printf("\n");  
  
printf("%s : ", product3);  
formatPrice(price3);  
printf("\n");  
  
return 0;  
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS  
✓ TERMINAL  
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc price.c -o price.exe  
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\price.exe  
Product Prices:  
Wireless Mouse : Rs1,599  
Keyboard : Rs899  
Headphones : Rs2,499  
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```

5) Refactor an existing function you wrote above to make it reusable for both product names and usernames (for example, a function that capitalizes the first letter of any string).

 Constraint: The refactored function should work for any string input, not just a specific use case.

Code-

```
#include <stdio.h>

#include <ctype.h>

void capitalizeFirstLetter(char str[])
{
    if (str[0] != '\0') // Check string is not empty
    {
        str[0] = toupper(str[0]);
    }
}

int main()
{
    char product[50];
    char username[50];

    printf("Enter product name: ");
    scanf("%s", product);
```

```
    capitalizeFirstLetter(product);

    printf("Product Name: %s\n", product);

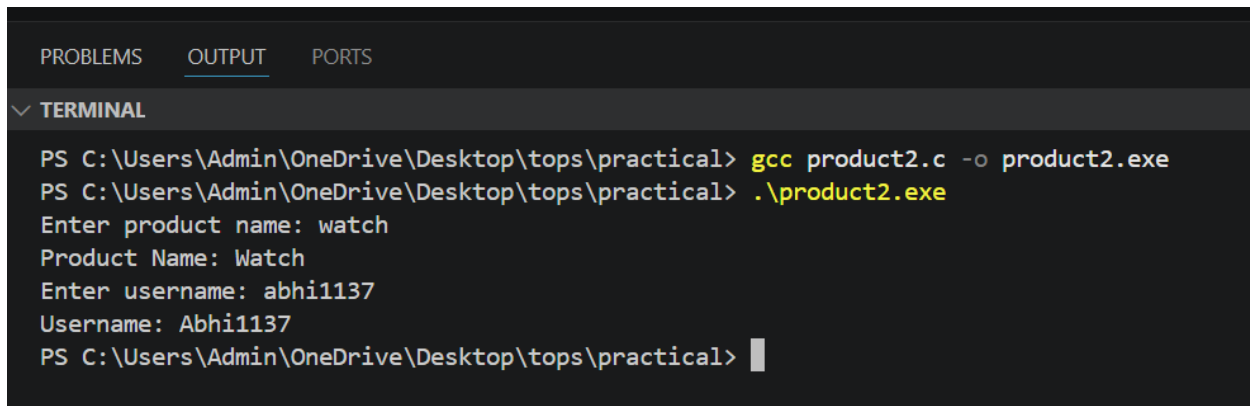
    printf("Enter username: ");
    scanf("%s", username);

    capitalizeFirstLetter(username);

    printf("Username: %s\n", username);

    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS
✓ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc product2.c -o product2.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\product2.exe
Enter product name: watch
Product Name: Watch
Enter username: abhi1137
Username: Abhi1137
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```

